

2022

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : CC-7P

(Operating Systems)

Full Marks : 30

Batch - 1

Answer *any one* question.

1. Write a shell script that prompts the user to input name of a file or directory and reports if it is a regular file or directory. If file, print the lengths of the smallest four lines and if directory, print the number of files it contains. Show both the cases.

2. Write a shell script that sorts list of integers using selection sort. Report if input/dataset is empty.

3. Write a shell script for menu based system to insert records for employees with employee id, name, designation and salary in a data file, also display records when necessary. Display salary of the employee asked using script. Check if record file is empty, create a new file and start with insertion job using your script.

4. Write a shell script that requires user inputs (consider, command line arguments only) and computes average value of those integers whose input value is greater than or equal to 10. Report if no input is passed to the script.
For example :
Input = 5 2 30 1 19 4 2
Output = 24.5

5. Write a shell script to count how many numbers are divisible by five and ten individually and together within user inputted range of integers.
For example :
Lower range = 6
Upper range = 16
How many numbers are divisible by five = 2
How many numbers are divisible by ten = 1
How many numbers are divisible by both five and ten = 1.
 6. Write a shell script that reads name of a text file and display frequency of each of the vowels in every line.
 7. Write a shell script to find the second largest element from a list of numbers. Report if input/dataset is empty.
 8. Write a shell script that multiply two 2-dimensional matrices and display result. Check if the multiplication is possible/compatible or not and report accordingly.
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